
UNIT 9 MONETARY POLICY IN INDIA: TRANSMISSION MECHANISM

Structure

- 9.0 Objectives
- 9.1 Introduction
- 9.2 Theoretical Framework of Monetary Policy Transmission
 - 9.2.1 Conceptual Overview
 - 9.2.2 Role of Financial Intermediaries
 - 9.2.3 Channels of Transmission
- 9.3 Financial Intermediaries and Monetary Policy Transmission
 - 9.3.1 Commercial Banks
 - 9.3.2 Non-Banking Financial Institutions
 - 9.3.3 Capital Markets
- 9.4 Credit Channel of Monetary Policy Transmission
 - 9.4.1 Mechanisms of Credit Provision
 - 9.4.2 Impact on Borrowing Costs
 - 9.4.3 Credit Availability and Investment
- 9.5 Interest Rate Channel of Monetary Policy Transmission
 - 9.5.1 Influences on Interest Rates
 - 9.5.2 Effects on Consumption and Investment
 - 9.5.3 Yield Curve Dynamics
- 9.6 Asset Price Channel of Monetary Policy Transmission
 - 9.6.1 Impact on Asset Prices
 - 9.6.2 Wealth Effects and Consumer Spending
 - 9.6.3 Implications for Financial Stability
- 9.7 Exchange Rate Channel of Monetary Policy Transmission
 - 9.7.1 Exchange Rate Dynamics
 - 9.7.2 Trade Balance and Competitiveness
 - 9.7.3 External Sector
- 9.8 Effectiveness and Challenges of Monetary Policy Transmission
 - 9.8.1 Policy Effectiveness Assessment
 - 9.8.2 Limitations and Constraints
 - 9.8.3 Adaptation to Changing Economic Conditions
 - 9.8.4 Coordination with Fiscal Policy
- 9.9 Let Us Sum Up
- 9.10 Key Words

9.11 Some Useful Books

9.12 Answers/Hints to Check Your Progress Exercises

9.13 Terminal Questions

9.0 OBJECTIVES

After going through this unit, you will be able to :

- Introduce the theoretical framework of monetary policy transmission;
- Understand the role of financial intermediaries, including commercial banks, non-banking financial institutions, and capital markets;
- Examine of the credit channel's mechanisms, impact on borrowing costs, and credit availability for investment;
- Analyse the interest rate channel's influences on interest rates, consumption, investment, and yield curve dynamics;
- Explore the asset price channel's impact on asset prices, wealth effects, and implications for financial stability; and
- Discuss the dynamics of the exchange rate channel, its effects on trade balance, competitiveness, and the external sector.

9.1 INTRODUCTION

In units 7 and 8 of this block, we discussed about Central bank, its role in monetary policy and its role as regulator of the banking system. We also discussed about objectives of monetary policy and various instruments through which objectives are achieved. This unit delves into the intricate mechanisms of monetary policy transmission, aiming to elucidate its theoretical framework and practical implications. By exploring the roles of financial intermediaries and various transmission channels, including credit, interest rates, asset prices, and exchange rates, learners gain a comprehensive understanding of how central bank actions reverberate through the economy. Through empirical evidence and analysis, the unit also examines the effectiveness and challenges inherent in monetary policy transmission.

9.2 THEORETICAL FRAMEWORK OF MONETARY POLICY TRANSMISSION

9.2.1 Conceptual Overview

Monetary policy framework refers to the set of principles, strategies, and tools used by a central bank to manage the nation's money supply, interest rates, and overall financial conditions to achieve specific economic goals. It provides a framework within which the central bank makes decisions regarding its monetary policy stance. Key components of Monetary

framework include Goals of monetary policy, targets and instruments. Figure 9.1 shows the schematic representation of the monetary policy framework that is followed around the globe by the Central banks.

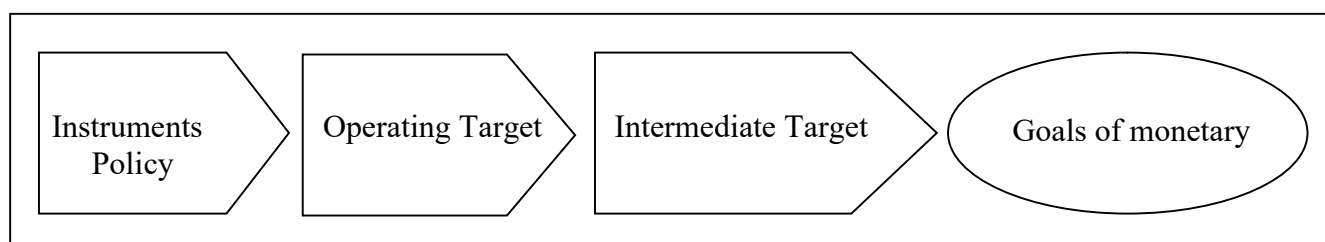


Fig. 9.1: A Schematic Representation of a Monetary Policy Framework

The evolution of India's monetary policy framework reflects shifts in economic priorities and global integration. From a focus on maintaining sterling parity to the adoption of monetary targeting and multiple indicators approaches, India has gradually adapted its policy framework to changing economic conditions.

Instruments of Monetary Policy

There are several direct and indirect instruments that are used for implementing monetary policy.

- 1) **Repo Rate:** The interest rate at which the Reserve Bank provides liquidity under the liquidity adjustment facility (LAF) to all LAF participants against the collateral of government and other approved securities.
- 2) **Standing Deposit Facility (SDF) Rate:** The rate at which the Reserve Bank accepts uncollateralised deposits, on an overnight basis, from all LAF participants. The SDF is also a financial stability tool in addition to its role in liquidity management. The SDF rate is placed at 25 basis points below the policy repo rate. With introduction of SDF in April 2022, the SDF rate replaced the fixed reverse repo rate as the floor of the LAF corridor.
- 3) **Marginal Standing Facility (MSF) Rate:** The penal rate at which banks can borrow, on an overnight basis, from the Reserve Bank by dipping into their Statutory Liquidity Ratio (SLR) portfolio up to a predefined limit (2 per cent). This provides a safety valve against unanticipated liquidity shocks to the banking system. The MSF rate is placed at 25 basis points above the policy repo rate.
- 4) **Liquidity Adjustment Facility (LAF):** The LAF refers to the Reserve Bank's operations through which it injects/absorbs liquidity into/from the banking system. It consists of overnight as well as term repo/reverse repos (fixed as well as variable rates), SDF and MSF. Apart from LAF, instruments of liquidity management include outright open market

operations (OMOs), forex swaps and market stabilisation scheme (MSS).

- 5) **LAF Corridor:** The LAF corridor has the marginal standing facility (MSF) rate as its upper bound (ceiling) and the standing deposit facility (SDF) rate as the lower bound (floor), with the policy repo rate in the middle of the corridor.
- 6) **Main Liquidity Management Tool:** A 14-day term repo/reverse repo auction operation at a variable rate conducted to coincide with the cash reserve ratio (CRR) maintenance cycle is the main liquidity management tool for managing frictional liquidity requirements.
- 7) **Fine Tuning Operations:** The main liquidity operation is supported by fine-tuning operations, overnight and/or longer tenor, to tide over any unanticipated liquidity changes during the reserve maintenance period. In addition, the Reserve Bank conducts, if needed, longer-term variable rate repo/reverse repo auctions of more than 14 days.
- 8) **Reverse Repo Rate:** The interest rate at which the Reserve Bank absorbs liquidity from banks against the collateral of eligible government securities under the LAF. Following the introduction of SDF, the fixed rate reverse repo operations will be at the discretion of the RBI for purposes specified from time to time.
- 9) **Bank Rate:** The rate at which the Reserve Bank is ready to buy or rediscount bills of exchange or other commercial papers. The Bank Rate acts as the penal rate charged on banks for shortfalls in meeting their reserve requirements (cash reserve ratio and statutory liquidity ratio). The Bank Rate is published under Section 49 of the RBI Act, 1934. This rate has been aligned with the MSF rate and, changes automatically as and when the MSF rate changes alongside policy repo rate changes.
- 10) **Cash Reserve Ratio (CRR):** The average daily balance that a bank is required to maintain with the Reserve Bank as a per cent of its net demand and time liabilities (NDTL) as on the last Friday of the second preceding fortnight that the Reserve Bank may notify from time to time in the Official Gazette.
- 11) **Statutory Liquidity Ratio (SLR):** Every bank shall maintain in India assets, the value of which shall not be less than such percentage of the total of its demand and time liabilities in India as on the last Friday of the second preceding fortnight, as the Reserve Bank may, by notification in the Official Gazette, specify from time to time and such assets shall be maintained as may be specified in such notification (typically in unencumbered government securities, cash and gold).
- 12) **Open Market Operations (OMOs):** These include outright purchase/sale of government securities by the Reserve Bank for injection/absorption of durable liquidity in the banking system.

The recent adoption of Flexible Inflation Targeting (FIT) framework, influenced by global best practices, underscores India's commitment to maintaining price stability while supporting economic growth. Under FIT, the repo rate serves as the policy rate, with the Reserve Bank of India (RBI) managing liquidity to achieve the desired interest rate outcomes.

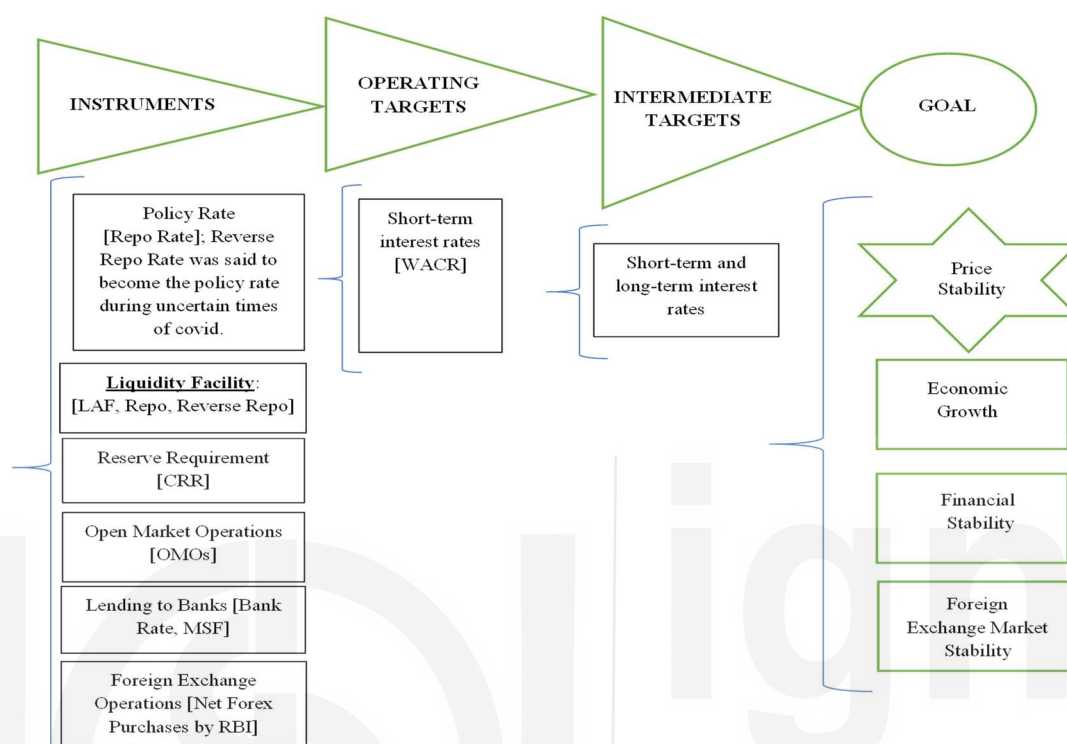


Fig. 9.2: Flexible Inflation Targeting Framework (FIT)

Figure 9.2 is a diagrammatical representation of the Flexible Inflation targeting framework followed by India. Further in the unit, we will take reference of the figure in the Indian context.

9.2.2 Role of Financial Intermediaries

As we have discussed in Unit 2 of this course, financial intermediaries serve as the linchpin of the financial system, facilitating the efficient allocation of capital between surplus and deficit units. They bridge the gap between savers and borrowers by pooling funds from individuals and institutions and deploying them into various investment avenues. By engaging in intermediation activities such as deposit-taking, lending, and risk management, these entities enhance liquidity, mitigate risk, and foster economic growth.

At the heart of their function lies the transformation of financial assets and liabilities, converting short-term liabilities (deposits) into long-term assets (loans and investments). This process not only ensures the efficient utilization of resources but also promotes stability within the financial system. Moreover, financial intermediaries play a pivotal role in reducing information asymmetry between borrowers and lenders, thereby fostering trust and confidence in the financial markets.

9.2.3 Channels of Transmission

The transmission of financial activities occurs through a myriad of channels, each playing a distinct role in shaping economic outcomes. These channels encompass various mechanisms through which changes in financial conditions reverberate throughout the economy, influencing consumption, investment, and overall economic activity.

One such channel is the interest rate channel, wherein changes in monetary policy rates affect borrowing costs, investment decisions, and aggregate demand. Additionally, the credit channel operates through alterations in the availability of credit, impacting the spending behaviour of households and businesses. Moreover, the asset price channel operates via changes in asset prices, influencing wealth effects and consumer spending patterns.

Furthermore, the exchange rate channel comes into play through fluctuations in currency values, affecting trade balances, inflation dynamics, and monetary policy effectiveness. Finally, the confidence channel operates through changes in investor sentiment and perceptions of risk, influencing asset prices and financial market stability.

Understanding these channels of transmission is essential for policymakers and market participants alike, as they navigate the intricacies of the financial system and anticipate the impact of various shocks and policy measures on economic outcomes.

In conclusion, the role of financial intermediaries and the channels of transmission are integral components of the financial system, shaping the allocation of resources, the dynamics of economic activity, and the stability of markets. By comprehensively exploring these concepts, this unit aims to provide readers with a deeper insight into the mechanisms that drive financial intermediation and the transmission of financial activities.

Check Your Progress 1

- 1) What are the key channels underlying the transmission of financial activity in an economy?
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- 2) How does the Flexible Inflation Targeting framework influence the monetary policy approach in India?
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9.3 FINANCIAL INTERMEDIARIES AND MONETARY POLICY TRANSMISSION

Financial intermediaries play a pivotal role in the transmission of monetary policy, acting as conduits through which central bank actions influence economic variables such as interest rates, credit availability, and investment decisions. This section delves into the mechanisms through which different types of financial intermediaries facilitate the transmission of monetary policy, including commercial banks, non-banking financial institutions (NBFIs), and capital markets.

9.3.1 Commercial Banks

Commercial banks are the cornerstone of the financial system, serving as primary intermediaries between depositors and borrowers. In the context of monetary policy transmission, commercial banks play a critical role in channelling central bank actions to the broader economy. When central banks adjust policy rates, such as the discount rate or the federal funds rate (the RBI repo and reverse repo rates in India), commercial banks respond by adjusting their own lending and deposit rates accordingly. These changes in interest rates influence the cost of borrowing for businesses and consumers, thereby impacting investment and consumption decisions. Moreover, changes in central bank policy can affect the availability of credit, as commercial banks adjust their lending standards and loan portfolios in response to changes in monetary conditions. We will discuss more about Commercial banks in Unit 15.

9.3.2 Non-Banking Financial Institutions

Non-banking financial institutions (NBFIs), including insurance companies, pension funds, and mutual funds, complement the role of commercial banks in the financial system. While they do not engage in traditional deposit-taking activities, NBFIs play a crucial role in intermediating funds between savers and borrowers. In the context of monetary policy transmission, NBFIs influence the availability and cost of credit through their participation in financial markets. For instance, changes in central bank policy rates can impact the returns on fixed-income securities held by NBFIs, influencing their investment decisions and portfolio allocations. Additionally, NBFIs may engage in lending activities or provide liquidity through money market instruments, thereby transmitting monetary policy impulses to the broader economy. Non-banking financial intermediaries will be discussed in greater length in Unit 16.

9.3.3 Capital Markets

Capital markets serve as a vital channel for the transmission of monetary policy, providing avenues for the exchange of long-term securities such as stocks and bonds. Central bank actions can influence capital market conditions through changes in interest rates, liquidity provision, and signalling effects. When central banks adjust policy rates, they affect the

pricing of fixed-income securities, equity valuations, and the cost of capital for businesses. Moreover, central bank interventions in capital markets, such as open market operations and asset purchase programs, can influence market liquidity and risk appetite. These changes in capital market conditions impact investment decisions, asset prices, and overall economic activity, thereby transmitting monetary policy impulses throughout the economy. We will read more about Capital markets in Unit 12.

In summary, financial intermediaries play a crucial role in the transmission of monetary policy, channelling central bank actions to the broader economy through various mechanisms. By understanding the roles of commercial banks, non-banking financial institutions, and capital markets in the transmission process, policymakers and market participants can better anticipate the effects of monetary policy on economic variables and make informed decisions accordingly.

Check Your Progress 2

- 1) How do commercial banks contribute to the transmission of monetary policy in the economy?
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- 2) What role do non-banking financial institutions play in the transmission of monetary policy?
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- 3) How do central bank actions influence capital markets and their impact on the broader economy?
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9.4 CREDIT CHANNEL OF MONETARY POLICY TRANSMISSION

The credit channel of monetary policy transmission plays a pivotal role in the propagation of central bank actions throughout the economy, particularly influencing borrowing costs, credit availability, and investment decisions. This section explores the mechanisms through which the credit channel operates its impact on borrowing costs, and its implications for credit availability and investment.

9.4.1 Mechanisms of Credit Provision

The credit channel operates through various mechanisms by which changes in monetary policy conditions affect the provision of credit by financial intermediaries. When central banks adjust policy rates, commercial banks respond by recalibrating their lending rates and credit standards. Lowering policy rates stimulates borrowing activity by reducing the cost of funds for banks, which in turn lowers interest rates for borrowers. Conversely, tightening monetary policy leads to higher borrowing costs as banks adjust their lending rates upwards to maintain profitability and manage risk.

Additionally, changes in central bank liquidity provision and regulatory policies influence the overall availability of credit in the economy. For instance, central bank interventions in money markets and open market operations affect bank reserves and liquidity conditions, thereby influencing banks' ability to extend credit to households and businesses. Moreover, regulatory measures such as capital adequacy requirements and loan-to-value ratios impact banks' lending behaviour and the availability of credit across different sectors of the economy.

9.4.2 Impact on Borrowing Costs

The credit channel exerts a direct impact on borrowing costs, influencing interest rates on loans and other forms of credit extended to households and businesses. When central banks lower policy rates, borrowing costs decline, stimulating demand for credit and encouraging investment and consumption. Conversely, raising policy rates increases borrowing costs, dampening credit demand and slowing economic activity.

Moreover, changes in central bank policy rates influence the pricing of fixed-income securities and debt instruments in capital markets, affecting yields on corporate bonds, mortgage-backed securities, and other financial assets. These changes in asset prices and market interest rates further influence borrowing costs for businesses and consumers, shaping their investment and financing decisions.

9.4.3 Credit Availability and Investment

The credit channel also influences credit availability and investment decisions by affecting banks' willingness to extend loans and the overall supply of credit in the economy. Lowering policy rates encourages banks to expand their lending activities, leading to increased credit availability and easier access to financing for households and businesses. This, in turn, stimulates investment in productive activities, such as business expansion, capital expenditure, and housing construction.

Conversely, tightening monetary policy reduces credit availability as banks become more risk-averse and tighten their lending standards. This constriction in credit supply can impede investment and economic growth, leading to a slowdown in economic activity and a decline in asset prices.

Moreover, changes in credit conditions and the availability of financing can amplify the effects of monetary policy on aggregate demand, investment, and employment dynamics.

In summary, the credit channel of monetary policy transmission plays a crucial role in shaping borrowing costs, credit availability, and investment decisions in the economy. By understanding the mechanisms through which changes in monetary policy conditions influence the provision of credit by financial intermediaries, policymakers and market participants can better anticipate the effects of monetary policy actions on economic variables and adjust their strategies accordingly.

Check Your Progress 3

- 1) How does the credit channel of monetary policy transmission influence borrowing costs in the economy?
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- 2) What are the mechanisms through which changes in central bank policy rates affect credit availability?
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- 3) How does the tightening of monetary policy impact investment decisions and credit availability according to the credit channel?
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9.5 INTEREST RATE CHANNEL OF MONETARY POLICY TRANSMISSION

The interest rate channel of monetary policy transmission serves as a critical mechanism through which central bank actions influence economic activity by affecting interest rates across various financial markets. This section examines the influences on interest rates, the effects on consumption and investment, and the dynamics of the yield curve within the context of the interest rate channel.

9.5.1 Influences on Interest Rates

Central bank policy actions, such as changes in the target federal funds rate or the discount rate, exert direct influence on short-term interest rates. When